

# Liam Moroy, PhD

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## SUMMARY

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Machine learning researcher and engineer specialising in deep generative models (diffusion, score-based, flow matching) for image generation, reconstruction, and enhancement. Experienced in designing task-specific neural architectures, training at scale on GPU / HPC clusters, and reducing the inference cost and latency of generative models. Peer-reviewed publications, with a track record of taking methods from research prototype through quantitative validation to deployed service.

## EXPERIENCE

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### AI Engineer / Research Associate, Generative AI for Inverse Problems

Dec. 2025 – Present

*Heriot-Watt University*

*Edinburgh, Scotland*

- Built diffusion and flow-matching generative models for image reconstruction and enhancement across multiple imaging modalities (nuclear Computed Tomography, non-conventional optics, medical imaging), adapting network architectures and training objectives to each modality and training at scale on an HPC cluster.
- Accelerated conditional generative-model sampling by up to 20 times (diffusion) over state-of-the-art baselines while preserving output quality, directly reducing inference latency and runtime cost.
- Deployed trained diffusion and flow-matching models as REST inference endpoints via FastAPI, exposing reconstruction and posterior sampling as a reproducible service usable by non-ML collaborators.
- Improved the posterior-sampling fidelity of generative models, producing samples more faithful to the true Bayesian posterior and yielding better reconstruction quality and better-calibrated uncertainty estimates.
- Collaborated within a multi-disciplinary team of statisticians, physicists, and engineers on high-impact scientific and medical imaging applications.

### Doctoral Researcher, Deep Learning for Tomographic Image Reconstruction

2022 – 2025

*ONERA, the French Aerospace Lab & Université Paris-Saclay*

*Palaiseau, France*

- Three years of research at France's national aerospace laboratory on recovering images from severely incomplete and noisy sensor data, a problem shared by optronic and remote-sensing systems operating under acquisition constraints.
- Designed, implemented and validated deep learning reconstruction methods for sparse-view computed tomography, taking methods from state-of-the-art review through prototype to quantitative validation.
- Built a rigorous benchmarking methodology to evaluate whether learned models genuinely recover the underlying signal rather than producing plausible but unfaithful outputs, a failure mode critical to operational trust. This analysis formed the basis of two peer-reviewed publications.
- Benchmarked a range of ODE and SDE solvers and sampling schedules to characterise the trade-off between inference cost and output quality.
- Presented results to technical audiences at international conferences and to non-specialist audiences through interactive scientific outreach.

### AI Intern, Deep Learning for Tomographic Reconstruction of Compressible Flows

Feb. 2022 – Aug. 2022

*ONERA, the French Aerospace Lab*

*Paris, France*

- Designed and evaluated generative models for tomographic reconstruction of compressible flows, enabling conditional generation from sparse and noisy measurements.

### AI Intern, Deep Learning for Nebula Removal in Astronomical Images

Jul. 2021 – Sep. 2021

*Astrophysics Laboratory of Bordeaux*

*Bordeaux, France*

- Built a deep learning model to clean astronomical images of background nebula, using synthetic-to-real transfer learning to work around noisy labels.

## EDUCATION

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### Université Paris-Saclay & ONERA, the French Aerospace Lab

Paris, France

*Ph.D. in Deep Learning for Inverse Problems in Imaging*

*2022 – 2025*

### ENSEIRB-MATMECA (Bordeaux INP) & University of Bordeaux

Bordeaux, France

*MSc Electrical Engineering + MSc Complex Systems (concurrent)*

*2020 – 2022*

### ENSEIRB-MATMECA (Bordeaux INP)

Bordeaux, France

*BSc Electronics Engineering*

*2017 – 2020*

## SELECTED PROJECTS

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### Generative Modelling of Implied Volatility Surfaces | *Diffusion, PyTorch*

- Score-based diffusion model (VP-SDE, attention U-Net) trained on ~3,500 daily SPY implied-volatility surfaces (2010–2023); no-arbitrage consistency emerges from data without explicit constraints.
- Frames daily surface calibration as posterior sampling under a diffusion prior; benchmarked ODE/SDE samplers (Euler, Heun, DPM-Solver).
- *Quantitative Finance, Generative AI, Diffusion Models.* [github.com/LiamMry/iv-surface-diffusion](https://github.com/LiamMry/iv-surface-diffusion)

### Retrieval-Augmented Generation (RAG) for Scientific Literature | *RAG, LLMs, Docker*

- End-to-end, fully self-hosted RAG system for querying arXiv papers: PDF parsing (GROBID), Sentence-Transformer embeddings, ChromaDB vector search, and local LLM inference (Ollama).
- Designed a modular, incrementally re-runnable pipeline (ingestion, chunking, embedding, indexing, retrieval, generation) for reproducible, low-cost deployment with no external API dependencies.
- *RAG, LLMs, Vector Search, Embeddings, Docker.* [github.com/LiamMry/Scientific-RAG-system](https://github.com/LiamMry/Scientific-RAG-system)

## RESEARCH ACTIVITIES

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### Selected Publications

- “Getting Guidance Weights Right in Diffusion and Flow-Matching Posterior Sampling”, to be submitted, ICLR 2027, [L. Moroy](#) et al.
- “On the Posterior Gap in Plug&Play Diffusion Methods for Sparse-View CT”, IEEE JSTSP 2026 Special Issue on High-Dimensional Imaging, [L. Moroy](#), [G. Bourmaud](#), [F. Champagnat](#), [J.-F. Giovannelli](#).
- “Evaluating the Posterior Sampling Ability of Plug&Play Diffusion Methods in Sparse-View CT”, ICASSP 2025, [L. Moroy](#), [G. Bourmaud](#), [F. Champagnat](#), [J.-F. Giovannelli](#).

### Selected Oral Presentations

- “Improving the Posterior Sampling of PnP Diffusion Models by Super-Calibration: application to Poisson inverse problem”, 7th SLSIP Workshop, May 2026, Krujë, Albania.
- “Evaluating the Posterior Sampling Ability of Plug&Play Diffusion Methods in Sparse-View CT”, GdR IASIS: Advances in Learning-Based Image Restoration, Dec 2024, Henri Poincaré Institute, Paris.

## TEACHING & KNOWLEDGE TRANSFER

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- **ENSEIRB-MATMECA Engineering School:** Teaching assistant, Generative modelling with diffusion models, MSc level (2024/25); Introduction to programming in C/C++, first year (2023/24 & 2024/25).
- **University of Bordeaux:** Teaching assistant, Introduction to noise and random signals, Master ISC (2023/24 & 2024/25).
- **Scientific outreach:** Built interactive in-browser demonstrations of deep learning imaging methods for technical and non-specialist audiences.

## TECHNICAL SKILLS

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**Programming:** Python, C, C++

**Generative Modeling:** Score-Based Generative Modeling, Diffusion Models, Flow-Matching, Rectified Flow-Matching, World Models, Autoregressive Modeling, Normalizing Flows, Variational Auto-Encoders

**Deep Learning & Vision:** Neural Network Architectures (CNNs, U-Net, Attention), Loss/Objective Design, Image Reconstruction, Image Restoration & Enhancement, Large-Scale Model Training & Evaluation

**LLMs & GenAI Systems:** Agentic AI, Retrieval-Augmented Generation (RAG), LLM Inference (Ollama), Embeddings, Vector Databases (ChromaDB), Sentence Transformers, Prompt Engineering, LoRA fine-tuning

**MLOps & Deployment:** FastAPI, REST APIs, Model Serving, Docker, Git/GitHub, Linux/Bash

**ML & Scientific Computing:** PyTorch, CUDA, GPU-Accelerated Computing, Slurm, HPC Clusters

**Research Methods & Domains:** Computer Vision, Image & Signal Processing, Plug-and-Play Methods, Bayesian Inference, Inverse Problems, Uncertainty Quantification, Experimental Design

**Languages:** French (Native), English (Fluent; TOEIC 950), Spanish (B1; Intermediate)